

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

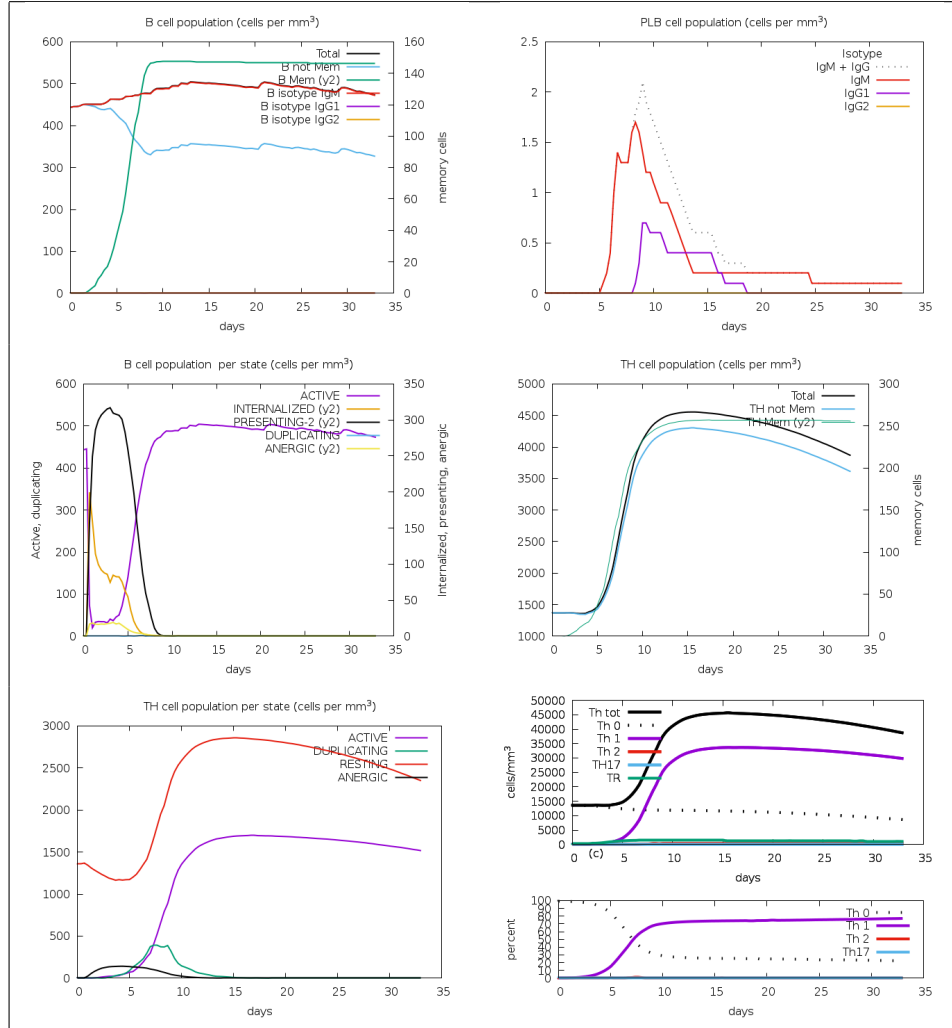


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

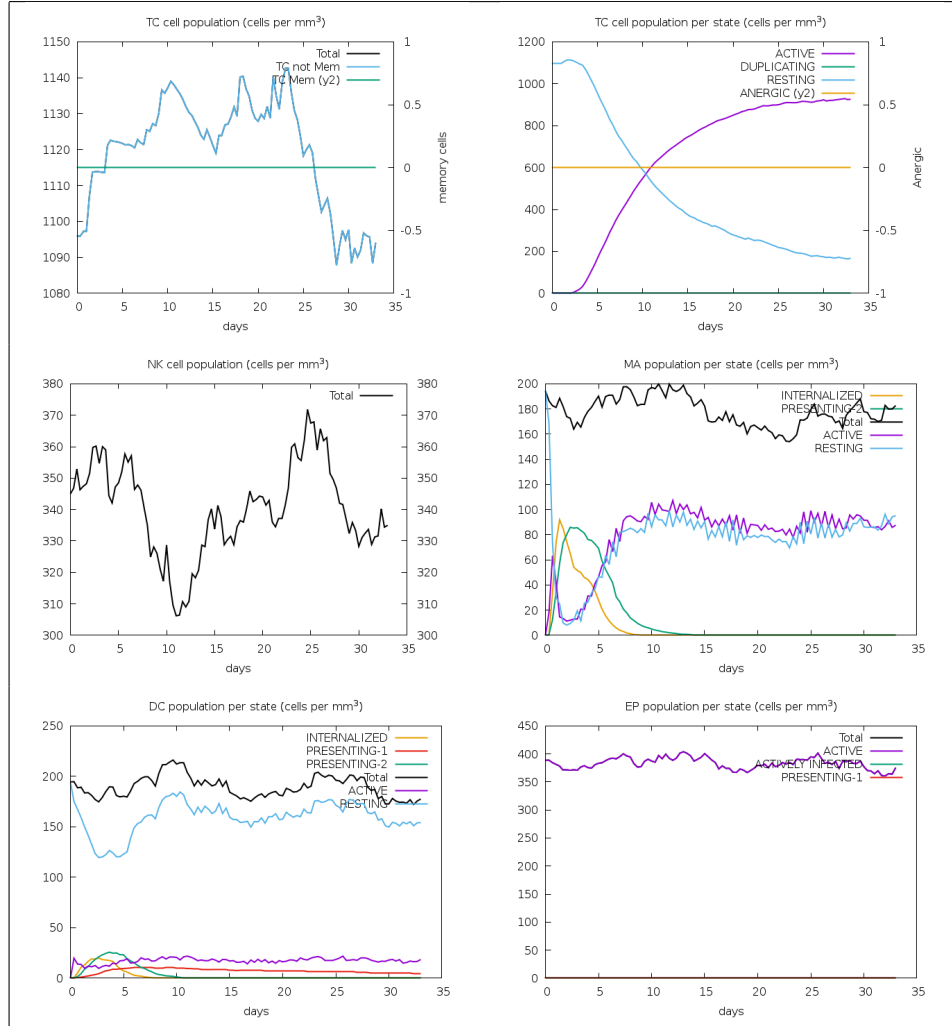


Figure 2: Legend: symbols as figure above.

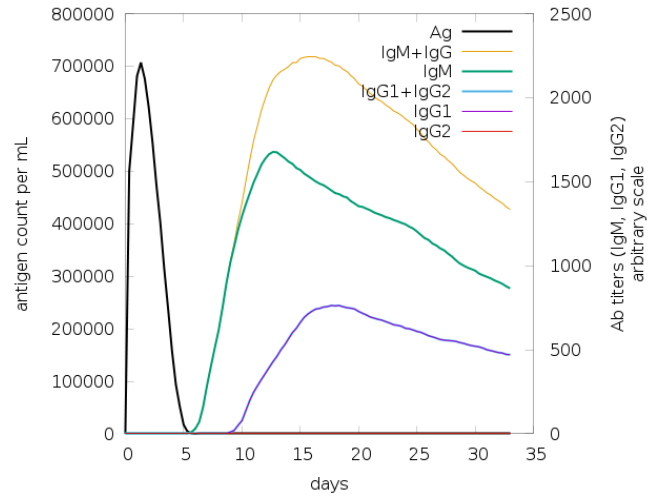


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

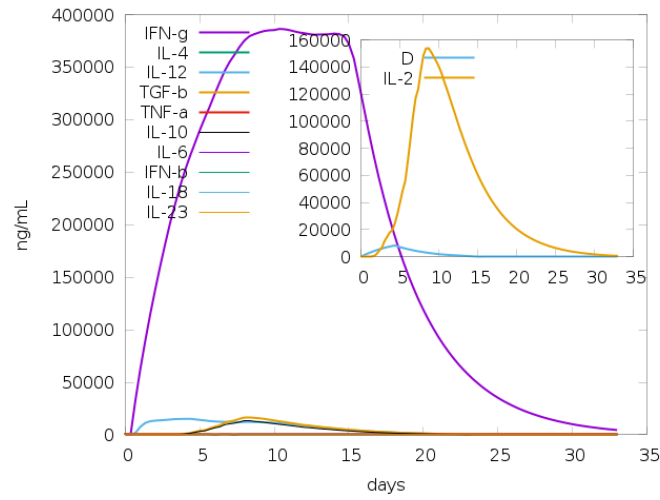


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

APPHALSGGSAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAAAKGPGTGPNGNL
GEKGDTSGPEGSGGSGPQRGGDNDHGRGRGRGRGRGGRRGPAGPGSSGSPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAG
AGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGAGGGAGAGAEEAAKRVLSVVLLGPGPGTMFEV
SVAFGPGPGACMLSAPLEKGPGPGPLQGGLTAYGPGPGFLDKGYTLTLPHTKAVYQPRGAFGGS

Epitopes of protein 0 -----

```

1]      pos=42 len=22      TQNQQTNQAGGEAPQPGDNSTV
2]      pos=70 len=6      GPGTGP
3]      pos=79 len=53      LGEKGDTSGPEGSGGSPQRRGDNHGRGRGRGRGRGGRRPGAPGGSGSGPRH
4]      pos=133 len=4      DGVR
5]      pos=149 len=66     GTHGGTGAGAGAGGAGAGGAGAGGGGAGAGGGAGGAGGAGGAGAGGGGAGAGGGAGGAGGAGAGGGAG

```

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 4 MHC I molecules

Read class I peptide list from file? NO

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

APPHALSGGSAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNQQTNQAGGEAPQPGDNSTVQEAATAKGGTGPNGNL
GEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRHRDGVRRPQKRPSCIGCKGTHGGTGAGAG
AGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEV
SVAFGPGPGACMLSAPLEKGPGPGLPQQQLTAYGPGPGFLDKGYTLTGPGPGLTPHTKAVYQPRGAFGGS

Epitopes of protein 0 -----

```
0]          pos= -1 score=1.000000 unnormalised=104.8410000000      non-binding event
```

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

5]	pos= 256	score=0.010087	unnormalised=0.8915600000	LEKGP GPGL
6]	pos= 284	score=0.078550	unnormalised=6.9425600000	YTLGP GPGL
7]	pos= 299	score=0.004408	unnormalised=0.3895600000	VYQPRGA FG
8]	pos= -1	score=0.748647	unnormalised=66.1680000000	non-binding event
